

TASK 25

FarmMarket RAG: AI-Powered Agricultural Market Intelligence and Decision Support System

Problem Statement

Agriculture is the primary source of income for millions of farmers, yet many continue to face financial losses due to a lack of timely and reliable market information. Critical data such as crop prices, market demand, weather forecasts, storage facilities, government schemes, and agricultural advisories are scattered across multiple sources, making it difficult for farmers to access and utilize effectively. As a result, farmers often rely on intermediaries or outdated information when making decisions about harvesting, storage, and selling their produce.

Existing agricultural information systems mainly provide static data and require users to manually search through multiple websites and reports. These systems do not offer personalized recommendations or intelligent analysis based on a farmer's specific crop, location, and market conditions. Consequently, farmers may sell their produce at lower prices, miss profitable market opportunities, or remain unaware of government support programs.

To address this challenge, the proposed FarmMarket RAG system leverages Retrieval-Augmented Generation and Large Language Models to create an intelligent agricultural decision-support platform. The system retrieves relevant information from agricultural reports, market databases, weather advisories, and government scheme documents, and generates accurate, context-aware recommendations in natural language. By providing real-time market insights, price trend analysis, government scheme awareness, and personalized selling strategies, the system aims to empower farmers with data-driven decision-making capabilities, improve profitability, and reduce dependence on intermediaries.

The proposed solution contributes to sustainable agricultural development by transforming fragmented agricultural data into actionable intelligence that can directly benefit farmers and agricultural stakeholders.